BLOOD DONOR SYSTEM PORTAL – EMERGENCY BLOOD DONOR COORDINATION SYSTEM

A PROJECT REPORT

Submitted by

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in partial fulfilment of the End Semester Examination for the course

CB23F35 –
LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

In

BACHELOR OF ENGINEERING
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NOVEMBER 2026

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Certified that this project report titled "**BLOOD DONOR SYSTEM PORTAL – EMERGENCY BLOOD DONOR COORDINATION SYSTEM**" is the bonafide work of Sahana A [2116240701451], who carried out the project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator during the academic year [2026-2027], under supervision.

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I, Sahana A, [2116240701451], hereby declare that the project report titled “Blood Donor System Portal – Emergency Blood Donor Coordination System”, submitted for the End Semester Examination of the course CB23F35– Low-Code Application Development using Zoho Creator, is a record of original work carried out by me. The Zoho Creator application described in this report was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature, code snippet or external resource used in this work has been duly acknowledged in the report.

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ABSTRACT

During emergencies, patients and their families often struggle to find suitable blood donors quickly. Donor contact information is scattered across phones, notebooks and messaging groups, and coordinating donors and requests manually takes time. Requests are not recorded, urgency is not tracked and completed donations are not documented.

The Blood Donor System Portal is a centralized blood donor coordination application developed using the Zoho Creator low-code platform. It consists of five linked forms – Donor_Registration, Blood_Request, Hospital_Registration, Donor_Match and Donation_History – supported by reports and a dashboard. Deluge scripting and workflows are designed to set a default request status of Pending, to shortlist potential donors by blood group, location and availability, to update match and request statuses and to maintain donation history.

The portal identifies potential donors for further verification only; it does not establish medical eligibility or clinical blood compatibility, which must be confirmed by qualified professionals. This report documents the requirements, design, implementation approach and test plan. Features are classified as designed, implemented and verified, or proposed for the future; screenshots and test results are to be added from the live application.

Keywords: Low-Code Development; Zoho Creator; Deluge Scripting; Blood Donor Management; Workflow Automation; Database Management.

PROJECT AT A GLANCE

Attribute	Details
Project title	Blood Donor System Portal – Emergency Blood Donor Coordination System
Domain	Web application development, database management, healthcare support system
Student name and register no.	SAHANA A,240701451
Development platform	Zoho Creator low-code platform
Scripting language	Deluge
Project type	Academic mini project / web-based application
Forms	5 forms: Donor_Registration, Blood_Request, Hospital_Registration, Donor_Match, Donation_History
Reports	8 proposed reports (Appendix B)
Dashboard	7 proposed indicators (Table 5.3)
Roles	Administrator; Donor / Requester (proposed)
Automation	Default status, donor matching, status updates, donation history update, optional notifications (designed)
External integrations	None assumed; SMS and e-mail at scale are future enhancements
Deployment status	Not evidenced; published URL and test evidence to be added

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update and Delete
DFD	Data Flow Diagram
ER	Entity–Relationship
GPS	Global Positioning System
KPI	Key Performance Indicator
RBAC	Role-Based Access Control
SMS	Short Message Service
UAT	User Acceptance Testing
UI	User Interface
URL	Uniform Resource Locator

CHAPTER 1
INTRODUCTION

1.1 Background

Blood is a life-saving resource that cannot be manufactured; it can only be obtained from willing human donors. In a medical emergency such as an accident, a surgery or a childbirth complication, a patient may need blood of a specific group within a very short time. Hospitals and blood banks maintain stocks, but a stock of a particular group can run out, and the family of the patient is then forced to search for a voluntary donor on their own.

In practice this search is informal. Relatives call friends, forward messages to groups and post requests on social media. Donor contact numbers are scattered across notebooks, phone contacts and chat groups; nobody knows with certainty who is currently available, which blood group they hold or how far they live from the hospital. Valuable time is lost in verifying and re-verifying this information.

The Blood Donor System Portal is an emergency blood donor coordination system developed on the Zoho Creator low-code platform. It brings donor records, blood requests, hospital details, donor matching and donation history into a single centralised application. It is a record-keeping and coordination tool: it helps an administrator to identify **potential** donors quickly and to track the progress of every request, while the medical decision about donor eligibility and blood compatibility remains with qualified healthcare professionals.

1.2 Problem Statement

During emergencies, patients and their families may face difficulties finding suitable blood donors quickly. Contact information may be scattered across different sources, and manually coordinating donors and blood requests can take time. The proposed Blood Donor System Portal provides a centralised platform to maintain donor records, register blood requests, manage hospital information, identify potential donor matches and maintain donation records. The specific problems addressed are:

- Donor details such as blood group, city and contact number are not held in one searchable place.
- Blood requests are communicated informally, so their urgency, status and outcome are not recorded.
- It is difficult to know which donors are currently available in the required city.
- The same donor may be contacted repeatedly for one request while other donors are never contacted.
- There is no record of completed donations, so repeated donation within a short time cannot be reviewed easily.
- Administrators have no consolidated view of pending and emergency requests.

1.3 Objectives

The main objective is to develop a centralised blood donor coordination portal that helps manage donor information, blood requests, hospitals, donor matching and donation history. The detailed objectives are explained below.

1. **Create a centralised database of blood donors** – all donor information is stored in one Zoho Creator form instead of notebooks and chat groups, so that it can be searched, filtered and reported.
2. **Maintain donor details** – name, contact number, blood group, city and availability are captured with mandatory-field validation so that the data is usable when an emergency arises.
3. **Allow users or administrators to register blood requests** – a structured request form records the patient, blood group, units, hospital, city, contact number and required date.
4. **Manage emergency and normal requests** – every request carries a type (Normal or Emergency) so that urgent cases can be filtered and handled first.
5. **Identify potential donors** – donors are shortlisted by blood group, location and availability, as a starting point for verification.
6. **Maintain hospital information** – hospital names, addresses, contact persons and blood bank information are held as reusable master data.
7. **Track donor matching and request status** – each match and each request moves through defined statuses so that progress is visible.
8. **Maintain donation history** – completed donations are recorded with date, hospital, units and the related request.
9. **Reduce manual coordination and improve accessibility** – reports and a dashboard give administrators one place to look instead of many.
10. **Provide an organised, user-friendly platform** – simple forms, filtered reports and a dashboard built with the Zoho Creator visual builders.

Objective ID	Objective	Intended evidence / output
OBJ-01	Centralised donor database	Donor_Registration form and All Donors report
OBJ-02	Maintain donor details	Mandatory fields, blood group and availability drop-downs
OBJ-03	Register blood requests	Blood_Request form with default status Pending
OBJ-04	Normal and emergency requests	Request Type field and Emergency Requests report
OBJ-05	Identify potential donors	Find Potential Donors workflow and Donor_Match records
OBJ-06	Maintain hospital information	Hospital_Registration form and Hospital List report
OBJ-07	Track matching and status	Match and request status fields; Pending Requests report
OBJ-08	Maintain donation history	Donation_History form and report
OBJ-09	Reduce manual coordination	Dashboard counters and filtered reports
OBJ-10	Organised, user-friendly platform	Navigation, forms, reports and dashboard

Table 1.1: Project Objectives and Measures

1.4 Scope of the Project

The scope covers donor registration, blood request registration, hospital registration, donor matching, donation history, reports, a dashboard and role-based access for an administrator and a donor/requester. Throughout this report, features are classified into three groups so that the report does not over-state the work done:

- **Designed features** – functions specified in this report (forms, workflows, reports, dashboard, roles). Their configuration in the live application must be confirmed by the student with screenshots.
- **Implemented and verified features** – only those for which execution evidence (screenshots, workflow execution history, test results) has been attached. None are claimed in this report until evidence is added.
- **Future enhancements** – SMS and e-mail notification at scale, a mobile application, GPS-based search, advanced analytics and hospital or blood bank integration.

The portal does not perform medical screening. It does not decide whether a donor is medically fit or whether a donor's blood is clinically compatible with a patient; these decisions must be taken by qualified professionals. The system is a forms, reports, workflows and Deluge scripting solution and does not use artificial intelligence or machine learning.

1.5 Justification for a Low-Code Approach

Zoho Creator provides visual form building, lookup relationships, reports, dashboards, workflow configuration and Deluge scripting within one platform. The Blood Donor System Portal is process-oriented: most requirements involve capturing data, validating it, filtering it, changing statuses and displaying counts. These are exactly the tasks that a low-code platform handles well. Building the same system with a traditional stack would require a database design, a back-end, a front-end, authentication and hosting before the first record could be saved. With Zoho Creator the effort is concentrated on the data model and the business rules. The capabilities and limitations of the platform follow the official documentation [1], [2].

1.6 Organisation of the Report

Chapter 2 studies the existing process and the proposed system, including a feasibility study. Chapter 3 presents stakeholders and requirements. Chapter 4 describes the architecture, data flow, database and workflow design. Chapter 5 documents the implementation of forms, reports, the dashboard, Deluge scripts and workflows. Chapter 6 presents the testing approach and test cases. Chapter 7 discusses deployment, security and governance. Chapter 8 reviews results, advantages and limitations. Chapter 9 concludes the report and lists future enhancements. The appendices contain a feature status checklist, report catalogue, sample Deluge scripts, a user manual, an evaluation rubric and a viva-voce record sheet.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing scenario, a family that needs blood phones relatives and friends, posts a message in a messaging group or on social media, and waits for responses. Each responder must be asked for their blood group, city and willingness, and the answers are not written down in any common place. Several people may offer help at the same time while others are never contacted. Once the need is met, nobody records who donated, when, or for which patient. Hospitals keep their own registers, which are not connected to the donors who volunteered.

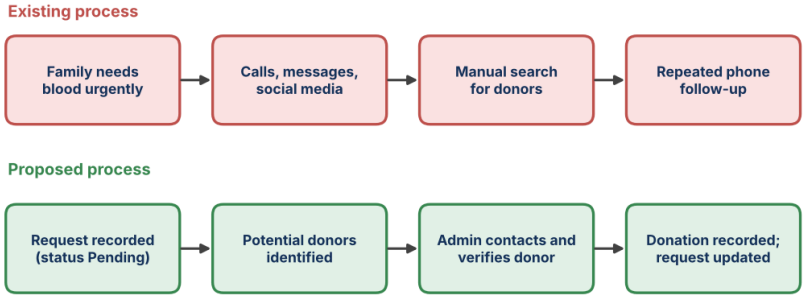


Figure 2.1: *Existing manual process compared with the proposed Blood Donor System Portal process*

Figure 2.1 contrasts the two processes. In the proposed process every step produces a record, so the status of a request is always known and the next action is clear to the administrator.

2.2 Limitations of the Existing System

- Donor information is fragmented across phones, notebooks and chat groups.
- There is no reliable way to filter donors by blood group, city and availability.
- Requests are not recorded, so urgency and status cannot be tracked.
- Time is lost in repeated phone calls and in verifying the same information again and again.
- Donation records are not maintained, so donation history cannot be reviewed.
- Information is not accessible to the people who need it at the moment they need it.
- Reports on requests and donors require manual consolidation and are quickly out of date.

2.3 Low-Code versus Traditional Development

Criterion	Traditional pro-code development	Low-code on Zoho Creator
Development effort	Custom UI, database, APIs and deployment setup are needed.	Visual form, report and dashboard builders with configured workflows.

Criterion	Traditional pro-code development	Low-code on Zoho Creator
Data capture	Forms and validation are coded individually.	Fields, drop-downs, lookups and mandatory rules are configured visually.
Automation	Business logic and schedulers are coded and hosted separately.	Workflow rules and Deluge scripts respond to form events.
Reporting	Queries and charts must be developed.	List reports, filters and dashboard components are built in.
Access control	Custom authentication and role management.	Users, profiles and permissions are configurable.
Hosting	Infrastructure and release process managed separately.	Hosting and publishing are provided by the platform.
Customisation	Broad control over architecture and UI.	Bounded by platform capabilities and plan limits.
Maintenance	Code-level maintenance and deployment discipline.	Routine changes are simpler, but scripts and permissions still need testing.

Table 2.1: Low-Code and Traditional Development Comparison

2.4 Literature Survey

The supplied project brief does not include a formal research-paper survey. This section therefore lists the authoritative documentation and standard textbooks on which the design is based, rather than claiming results from experiments that were not carried out.

S.No.	Source	Focus	Relevance
1	Zoho Creator Help documentation [1]	Forms, reports, pages and application building	Primary reference for platform capabilities
2	Zoho Deluge Help documentation [2]	Scripting language and task syntax	Validation, matching and automation logic
3	Elmasri and Navathe, Fundamentals of Database Systems [3]	Relational and ER modelling	Database and form design
4	Sommerville, Software Engineering [4]	Requirements, design and testing	Report structure, feasibility and test planning
5	Pressman, Software Engineering: A Practitioner's Approach [5]	Software process and quality	Requirement analysis and testing practice
6	MDN Web Docs [6]	Web application concepts	Browser-based client access
7	OWASP ASVS [7]	Application security verification	Access control and data protection checklist

Table 2.2: Literature Survey Sources

2.5 Proposed System

The proposed system centralises blood donor information in five linked Zoho Creator forms. When a blood request is submitted, its status is set to Pending automatically. The administrator runs the Find Potential Donors action, which shortlists donors whose blood group matches the request, who live in the required city and who are marked Available. Each shortlisted donor is recorded in the Donor_Match form so that the administrator can contact and verify them. When a donation takes place it is recorded in Donation_History and the

request status is updated. Reports and a dashboard give a live view of donors, requests, matches and donations.

Medical eligibility and clinical compatibility are outside the scope of the software. The matching result is a shortlist for further verification, not a medical clearance.

2.6 Feasibility Study

Technical feasibility

Zoho Creator supplies all the building blocks required: forms, lookups, drop-downs, reports, dashboards, workflows and Deluge. No additional hardware or software is required beyond a browser and an internet connection. The project is therefore technically feasible.

Economic feasibility

The development uses the Zoho Creator platform under the student or education access available to the developer, and no separate server, database licence or hosting cost is incurred. The long-term cost of production use depends on the Zoho plan chosen and should be reviewed by the organisation that adopts the portal.

Operational feasibility

The forms use plain labels and drop-down lists that an administrator can learn quickly. The workflow reduces the number of manual steps, so it is expected to fit the way a volunteer group or hospital desk already works.

Schedule feasibility

Because forms, reports and workflows are configured rather than coded from scratch, the application can be built within the duration of an academic semester.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Role	Responsibilities
Administrator	Manages donor records, blood requests and hospital details; runs donor matching; maintains donation history; views reports and the dashboard.
Donor	Registers personal and availability details; may update or view permitted information.
Requester	Submits a blood request for a patient and may view the status of the submitted request, as permitted.
Hospital contact	Provides hospital and blood bank information to be recorded by the administrator.

Table 3.1: Stakeholders and User Roles

3.2 Functional Requirements

ID	Requirement	Role(s)	Priority
FR-01	Register a donor with name, age, gender, blood group, phone, e-mail, city, address and availability.	Donor / Admin	High
FR-02	Generate a unique Donor ID and record the registration date automatically.	System	High
FR-03	Validate mandatory fields and reject incomplete donor or request records.	System	High
FR-04	Register a blood request with patient, blood group, units, hospital, city, contact number, type and required date.	Requester / Admin	High
FR-05	Set the status of every new request to Pending by default.	System	High
FR-06	Classify requests as Normal or Emergency and filter emergency requests.	Admin	High
FR-07	Register hospitals and store contact and blood bank information.	Admin	Medium
FR-08	Identify potential donors for a request by blood group, city and availability.	Admin / System	High
FR-09	Avoid creating duplicate matches for the same donor and request.	System	Medium
FR-10	Update match status and request status (Pending, In Progress, Fulfilled, Cancelled).	Admin	High
FR-11	Record completed donations linked to a donor and a request.	Admin	High
FR-12	Provide reports for donors, requests, matches, donations and hospitals.	Admin	Medium
FR-13	Provide a dashboard with counts of donors, requests, matches and donations.	Admin	Medium
FR-14	Send notifications to relevant users where configured.	System	Low
FR-15	Restrict functions according to the configured role permissions.	System	High

Table 3.2: Functional Requirements

3.3 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Data integrity	Lookups, drop-downs and validation should keep linked records consistent.
NFR-02	Security and privacy	Donor phone numbers and addresses are personal data and should be visible only to permitted roles.
NFR-03	Usability	Forms should be short and should work in a mobile browser.
NFR-04	Maintainability	Field names and statuses should be consistent so that Deluge scripts remain readable.
NFR-05	Auditability	Requests, matches and donations should retain traceable records.
NFR-06	Availability	Access depends on platform availability and network connectivity.
NFR-07	Performance	Reports should use filters so that only relevant records are loaded.

Table 3.3: Non-Functional Requirements

3.4 Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator low-code application platform
Scripting	Deluge for default values, validation, donor matching and notifications
Client device	Desktop, laptop or smartphone with a stable internet connection
Minimum hardware	Processor, memory and storage sufficient to run a current web browser (no special hardware)
Browser	Current version of Google Chrome, Microsoft Edge or Mozilla Firefox
Operating system	Any operating system that supports a current browser
Development tools	Zoho Creator form, report, dashboard and workflow builders
Optional integrations	SMS gateway or e-mail service (future); credentials and tests are required before use

Table 3.4: Hardware and Software Requirements

3.5 Use Case Model

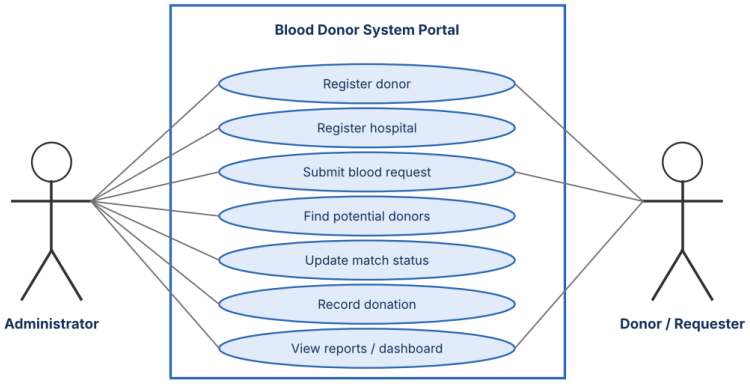


Figure 3.1: *Use case diagram of the Blood Donor System Portal*

The administrator interacts with every function of the portal, whereas the donor or requester is limited to registering information, submitting a request and viewing permitted information.

Use case	Precondition	Main flow	Result
Register donor	Donor has valid contact details	Open Donor_Registration, enter details, choose blood group and availability, submit	Donor record stored with ID and date
Submit blood request	Requester has patient details	Open Blood_Request, enter details and type, submit	Request stored with status Pending
Find potential donors	A Pending request exists	Admin selects request and runs the matching action	Donor_Match records created; request In Progress
Update status	Request or match exists	Admin edits status after contacting donors	Status updated and visible in reports
Record donation	Donor has donated	Admin enters donation details	Donation_History record; request updated
View dashboard	Admin is logged in	Open the dashboard page	Counts and reports displayed

Table 3.5: Use Case Descriptions

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

The architecture separates the user-facing layer, the automation layer and the stored application records. Users interact with Zoho Creator forms, reports and the dashboard. Deluge scripts and workflows enforce the process rules when a form is loaded, validated or submitted. The stored records feed the reports and dashboard counters.

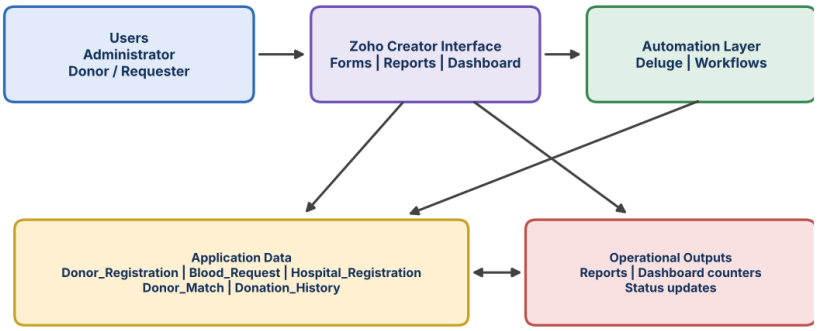


Figure 4.1: *Logical architecture of the Blood Donor System Portal*

4.2 Data Flow Design

The level 1 data flow diagram in Figure 4.2 shows five processes and four data stores. Donors and requesters supply data to the Manage Donors and Register Requests processes. The administrator drives donor matching, donation recording and reporting. Matching reads the donor and request stores and writes to the match store; a recorded donation writes to the donation history store.

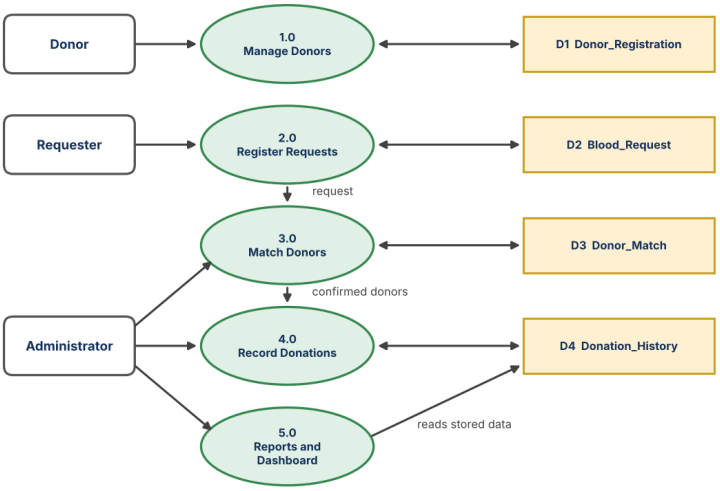


Figure 4.2: *Level 1 data flow diagram*

4.3 Database Design

The data model consists of five forms. Donor_Registration and Hospital_Registration hold master data. Blood_Request holds transactions that refer to a hospital. Donor_Match links one request to one donor, and Donation_History records completed donations linked to a donor and, optionally, to a request. In Zoho Creator these links are implemented as lookup fields. The relationships are shown in Figure 4.3.

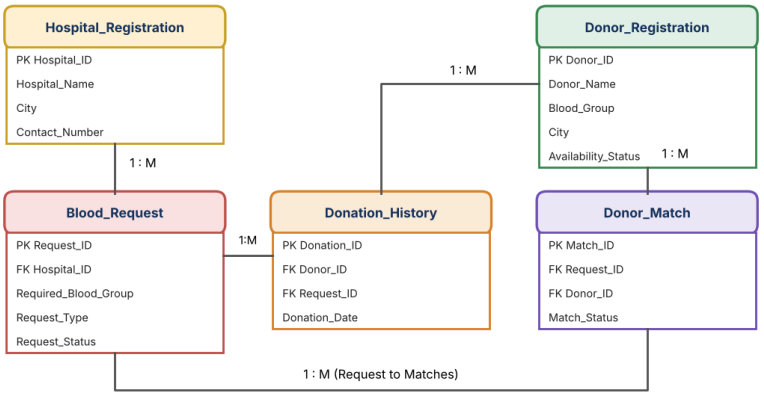


Figure 4.3: *Entity relationship diagram*

One hospital can have many requests; one request can have many matches; one donor can appear in many matches and many donations. The principal fields of each form are specified in Tables 4.1 to 4.5. Field types and link names are proposed and must be matched to the live application before the Deluge scripts are used.

Field	Type	Rule / notes
Donor_ID	Auto number	Unique identifier, system generated

Field	Type	Rule / notes
Donor_Name	Name / text	Mandatory
Age	Number	Mandatory; age limit to be defined by the organisation
Gender	Drop-down	Male, Female, Other
Blood_Group	Drop-down	Mandatory: A+, A-, B+, B-, AB+, AB-, O+, O-
Phone_Number	Phone	Mandatory; used for contact
Email	Email	Optional
City	Text / drop-down	Mandatory; used for location matching
Address	Address / text	Optional
Availability_Status	Drop-down	Available / Not Available; default Available
Registration_Date	Date	Defaults to the current date

Table 4.1: Field Specification – Donor_Registration

Field	Type	Rule / notes
Request_ID	Auto number	Unique identifier
Patient_Name	Text	Mandatory
Required_Blood_Group	Drop-down	Mandatory; same values as Donor Blood_Group
Units_Required	Number	Mandatory; greater than zero
Hospital_Name	Lookup (Hospital_Registration)	Mandatory
City	Text / drop-down	Mandatory
Contact_Number	Phone	Mandatory
Request_Type	Drop-down	Normal / Emergency
Required_Date	Date	Mandatory
Request_Status	Drop-down	Pending / In Progress / Fulfilled / Cancelled; default Pending

Table 4.2: Field Specification – Blood_Request

Field	Type	Rule / notes
Hospital_ID	Auto number	Unique identifier
Hospital_Name	Text	Mandatory; unique
Hospital_Address	Address	Optional
City	Text / drop-down	Mandatory
Contact_Person	Text	Optional
Contact_Number	Phone	Mandatory
Email	Email	Optional
Blood_Bank_Information	Multi-line text	Available blood groups or blood bank details

Table 4.3: Field Specification – Hospital_Registration

Field	Type	Rule / notes
Match_ID	Auto number	Unique identifier
Request_ID	Lookup (Blood_Request)	Mandatory
Donor_ID	Lookup (Donor_Registration)	Mandatory
Blood_Group, City	Text	Copied from the donor for convenience

Field	Type	Rule / notes
Contact details	Phone	Copied from the donor
Match_Status	Drop-down	Potential / Contacted / Confirmed / Declined / Donated

Table 4.4: Field Specification – Donor_Match

Field	Type	Rule / notes
Donation_ID	Auto number	Unique identifier
Donor_ID / Donor_Name	Lookup	Mandatory
Blood_Group	Drop-down	Mandatory
Donation_Date	Date	Mandatory
Hospital_Name	Lookup (Hospital_Registration)	Mandatory
Units_Donated	Number	Mandatory
Related_Request_ID	Lookup (Blood_Request)	Optional
Donation_Status	Drop-down	Completed / Cancelled

Table 4.5: Field Specification – Donation_History

4.4 Process and Workflow Design

Figure 4.4 shows the overall workflow. Donor and hospital records are created first. A blood request is then submitted and its status becomes Pending. The administrator runs the matching action; if matches are found the administrator contacts and verifies the donors. If a donor is confirmed and found eligible by the responsible professionals, the donation is recorded and the request is updated; otherwise the next donor is tried. If no match is found, the request remains Pending.

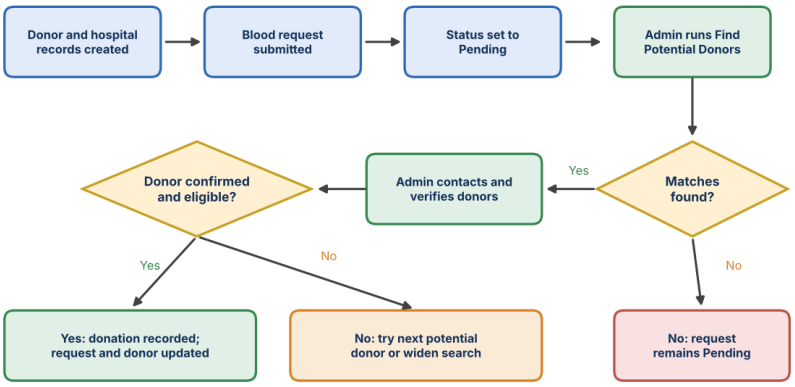


Figure 4.4: *Overall system workflow*

The status of a request is controlled through its Request_Status field, as shown in Figure 4.5. A request is created as Pending, becomes In Progress when potential donors have been identified, and becomes Fulfilled when the required units have been arranged. A Pending or

In Progress request can be Cancelled. The lifecycle is enforced through the drop-down values and the workflows; Zoho Creator Blueprint is not assumed.

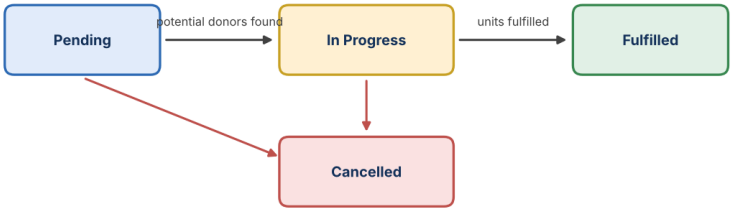


Figure 4.5: *Blood request status lifecycle*

4.5 User Interface and Navigation Design

The application is organised around the tasks of its users. The administrator menu lists the five forms and their reports, together with the dashboard page. A donor or requester sees only the registration or request form that has been made available to them. Status values use the same wording on forms, reports and the dashboard so that users do not need to learn different terms. Emergency requests are filtered into a dedicated report so that they are not hidden among normal requests.

4.6 Access Control Design

Role	Typical permitted actions	Restricted actions
Administrator	Create, edit and view donors, requests, hospitals, matches and donations; run matching; view the dashboard and all reports.	Privileged actions should be limited and reviewed periodically.
Donor / Requester	Submit own registration or request; view permitted information.	Cannot edit other users' records, run matching, or view the full donor list.

Table 4.6: Access Control Design

Note: the roles above are proposed. The report does not claim that portal access or role-based permissions are fully configured until they are verified in the live application.

CHAPTER 5
IMPLEMENTATION

Implementation status: this chapter describes the application as designed. Screenshots of the live Zoho Creator application are to be inserted at the placeholders, and a feature should be described as implemented and verified only after such evidence is attached.

5.1 Application Setup

The application "Blood Donor System Portal" is built on Zoho Creator using the visual builders for forms, reports, dashboards and workflows. A consistent naming scheme is followed (for example Donor_Registration, Blood_Request) because Deluge scripts refer to form and field link names. Before the workflows are published, the developer verifies the link names, the lookup targets, the mandatory fields and the order in which workflows run.

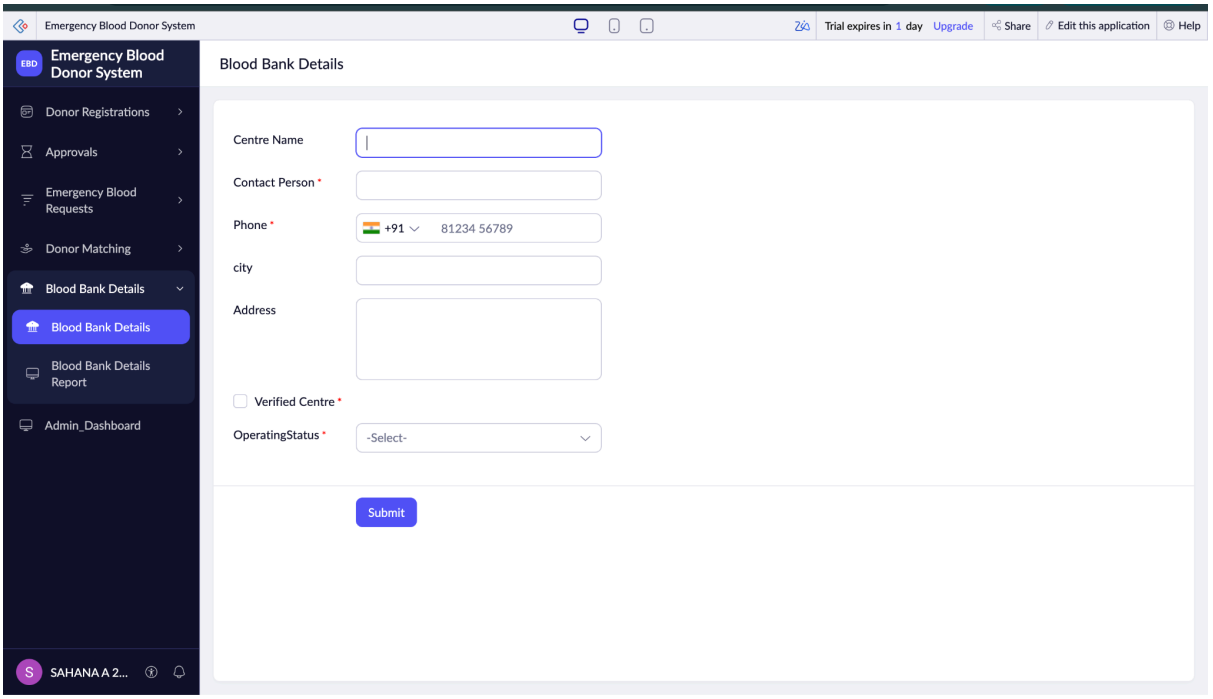


Figure 5.1: *Zoho Creator application home page*

5.2 Forms and Data Capture

No.	Form	Purpose and principal fields
1	Donor_Registration	Register and maintain donors: ID, name, age, gender, blood group, phone, e-mail, city, address, availability, registration date.
2	Blood_Request	Record blood requirements: ID, patient, blood group, units, hospital, city, contact, type, required date, status.
3	Hospital_Registration	Store hospital details: ID, name, address, city, contact person, contact number, e-mail, blood bank information.
4	Donor_Match	Record shortlisted donors for a request, with match status.
5	Donation_History	Record completed donations: ID, donor, blood group, date, hospital, units, related request, status.

Table 5.1: Forms Designed for the Application

5.2.1 Module 1 – Donor Registration (Donor_Registration)

This module registers and maintains blood donor information. The form captures identity, contact, blood group, city and availability. Blood Group and Availability Status are drop-down fields so that values are consistent, which is essential because the matching logic compares them exactly. Registration Date defaults to the current date and Donor ID is generated automatically. The Availability Status can be changed whenever the donor's situation chang

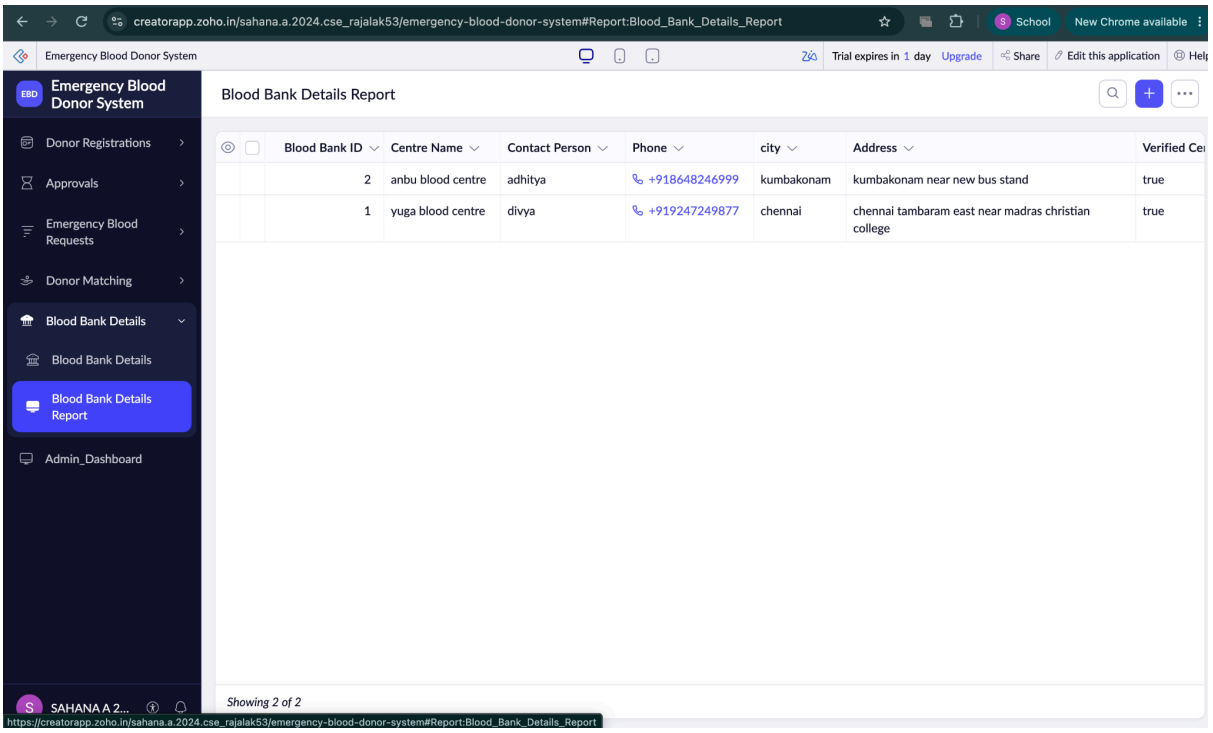


Figure 5.2: *Donor Registration form*

5.2.2 Module 2 – Blood Request (Blood_Request)

This module records blood requirements from patients, hospitals or requesters. The Hospital Name is chosen from the Hospital_Registration form through a lookup. Request Type distinguishes Normal from Emergency requests, and Request Status is set to Pending when the record is created. The administrator updates the status as the request progresses.

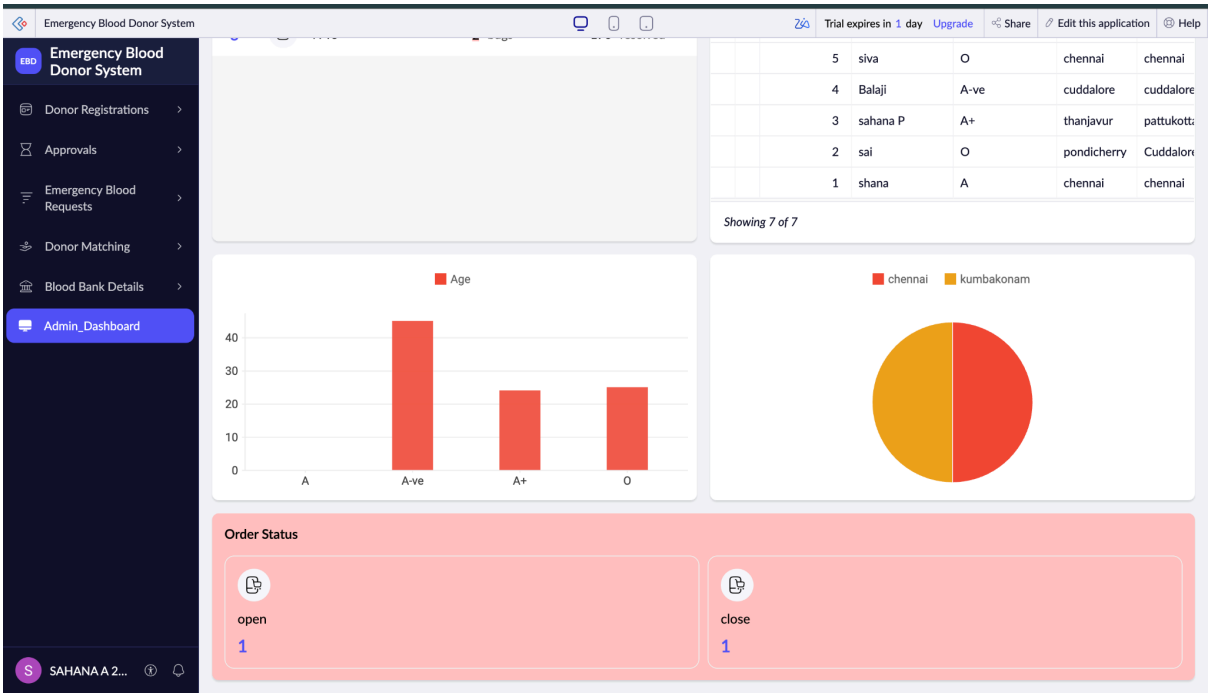


Figure 5.3: *Blood Request form*

5.2.3 Module 3 – Hospital Registration (Hospital_Registration)

This module stores hospital details and contact information so that they can be selected in requests and donations without re-typing. The Blood Bank Information field records the blood groups or blood bank details that the hospital has declared.

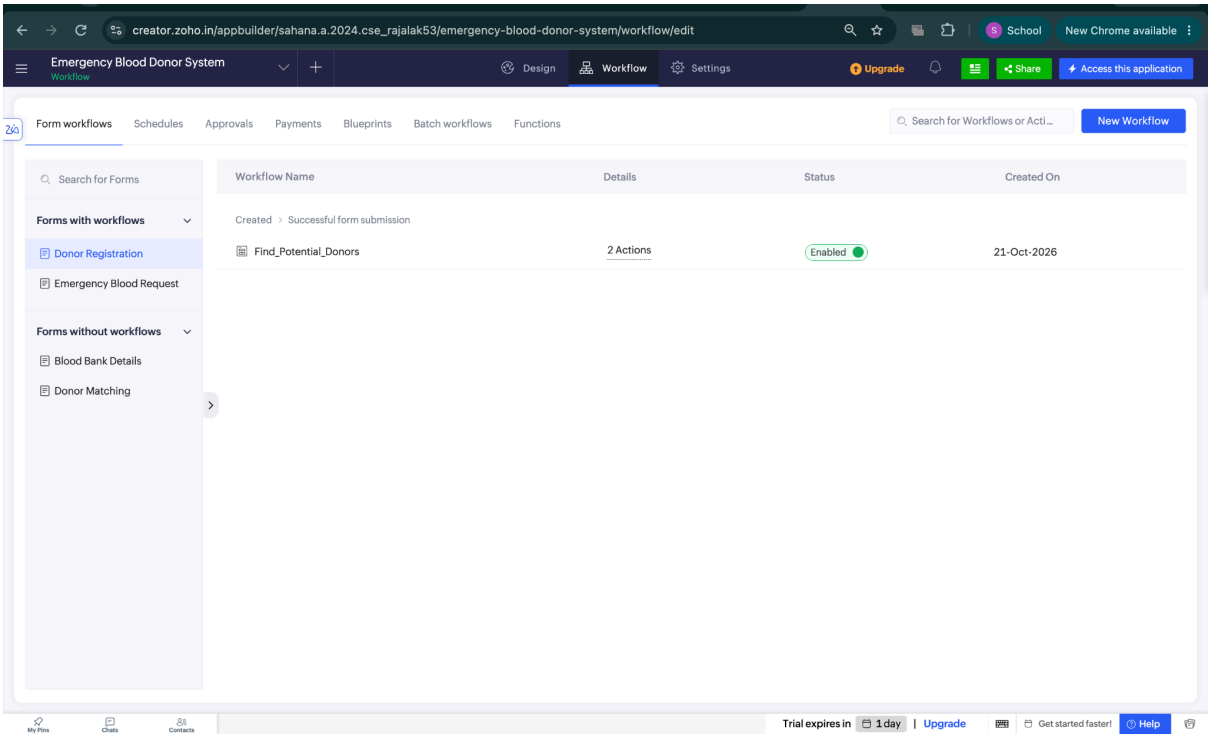


Figure 5.4: *Hospital Registration form*

5.2.4 Module 4 – Donor Matching (Donor_Match)

This module identifies potential donors for a registered blood request. The matching considers the required blood group, the donor's city, the donor's availability, the donor's contact details and the request status. The result is a shortlist of potential donors for further verification. **It does not independently establish medical eligibility or clinical blood compatibility**; these must be confirmed by qualified professionals.

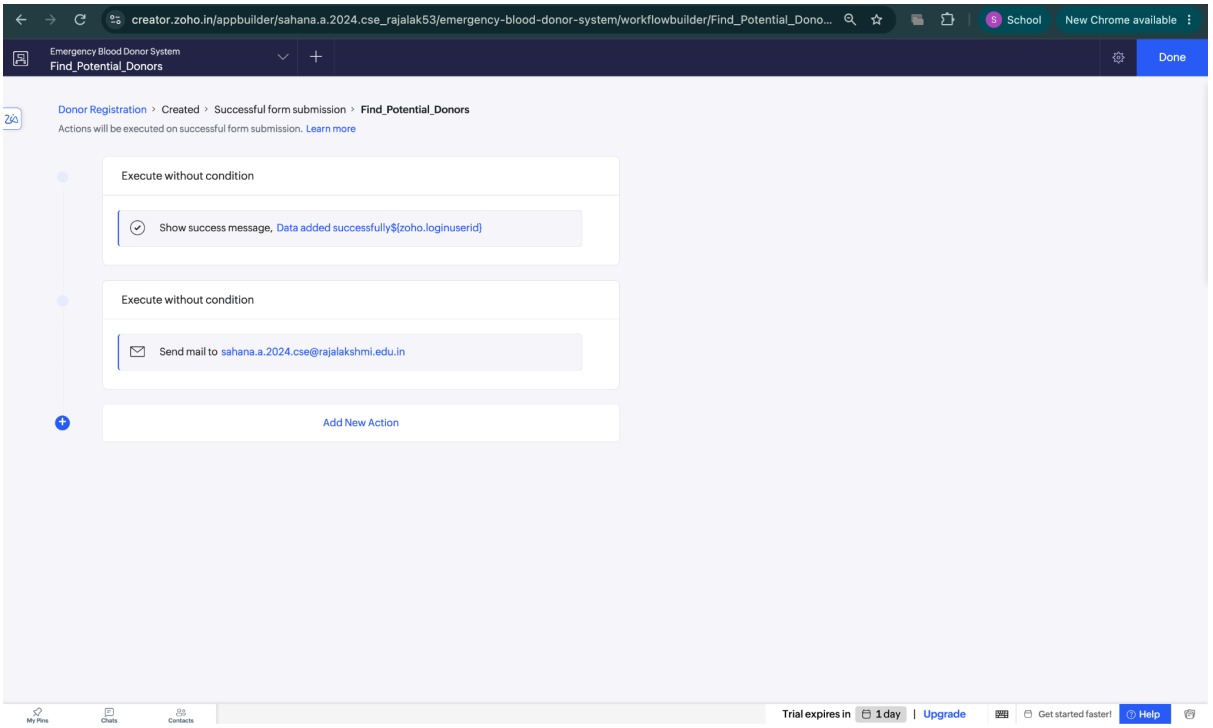


Figure 5.5: *Donor Match form*

5.2.5 Module 5 – Donation History (Donation_History)

This module maintains records of completed donations, including the donor, blood group, date, hospital, units donated, the related request and the donation status. The history supports review of how often a donor has donated and shows how each request was fulfilled.

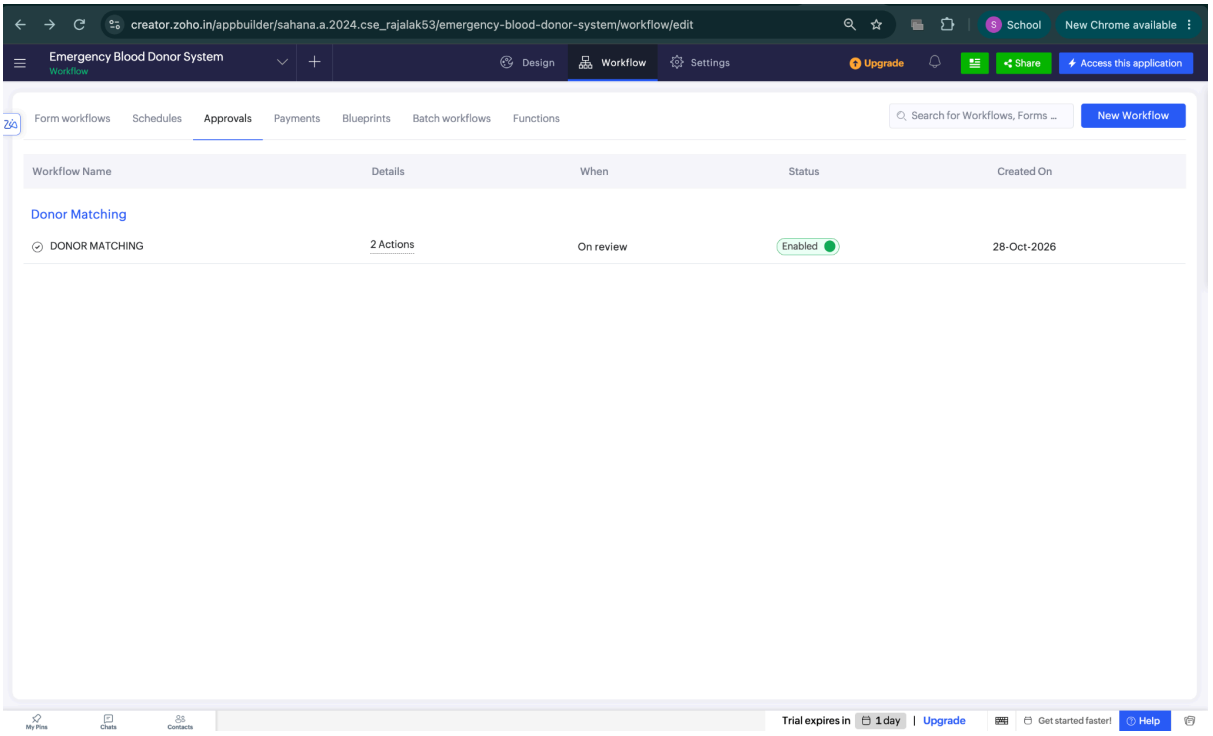


Figure 5.6: *Donation History form*

5.3 Reports

Reports provide tabular views of the stored records, filtered to the needs of the administrator. The proposed reports are listed in Table 5.2.

No.	Report	Use
1	All Donors	Review all registered donors, blood groups and cities.
2	Available Donors	Donors whose availability status is Available.
3	All Blood Requests	Review all requests with type and status.
4	Pending Requests	Requests whose status is Pending, awaiting action.
5	Emergency Requests	Requests whose type is Emergency.
6	Donor Matches	Shortlisted donors per request and match status.
7	Donation History	Completed donations with date, hospital and units.
8	Hospital List	Registered hospitals and contact details.

Table 5.2: Reports Designed for the Application

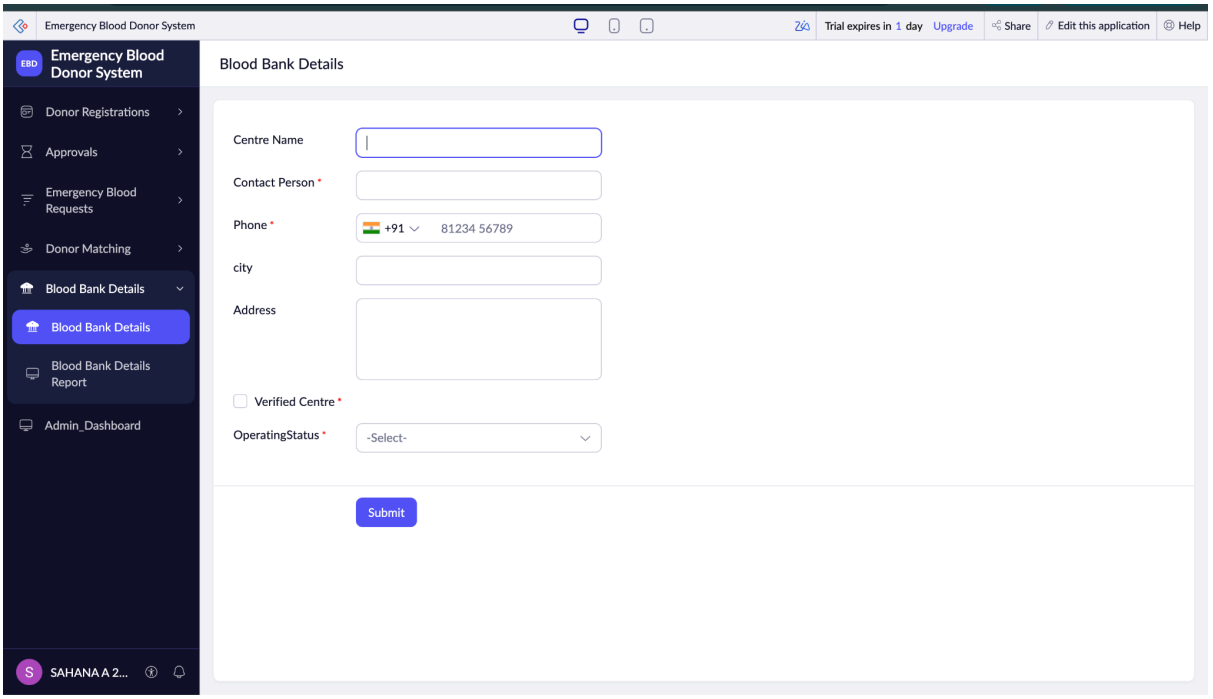


Figure 5.10: *Reports page*

5.4 Dashboard

The dashboard gives the administrator a single page that summarises the state of the system. Each counter is a count of the records of a form that satisfy a condition. The proposed components are listed in Table 5.3.

No.	Dashboard item	Meaning and source
1	Total Donors	Count of Donor_Registration records
2	Available Donors	Donors with Availability_Status = Available
3	Total Blood Requests	Count of Blood_Request records
4	Pending Requests	Requests with Request_Status = Pending
5	Emergency Requests	Requests with Request_Type = Emergency
6	Successful Matches	Donor_Match records with status Confirmed or Donated
7	Completed Donations	Donation_History records with status Completed

Table 5.3: Dashboard Items

These counters help the administrator to see at a glance how many requests are waiting, how many are emergencies and whether enough donors are available, so that attention is directed to the cases that need it.

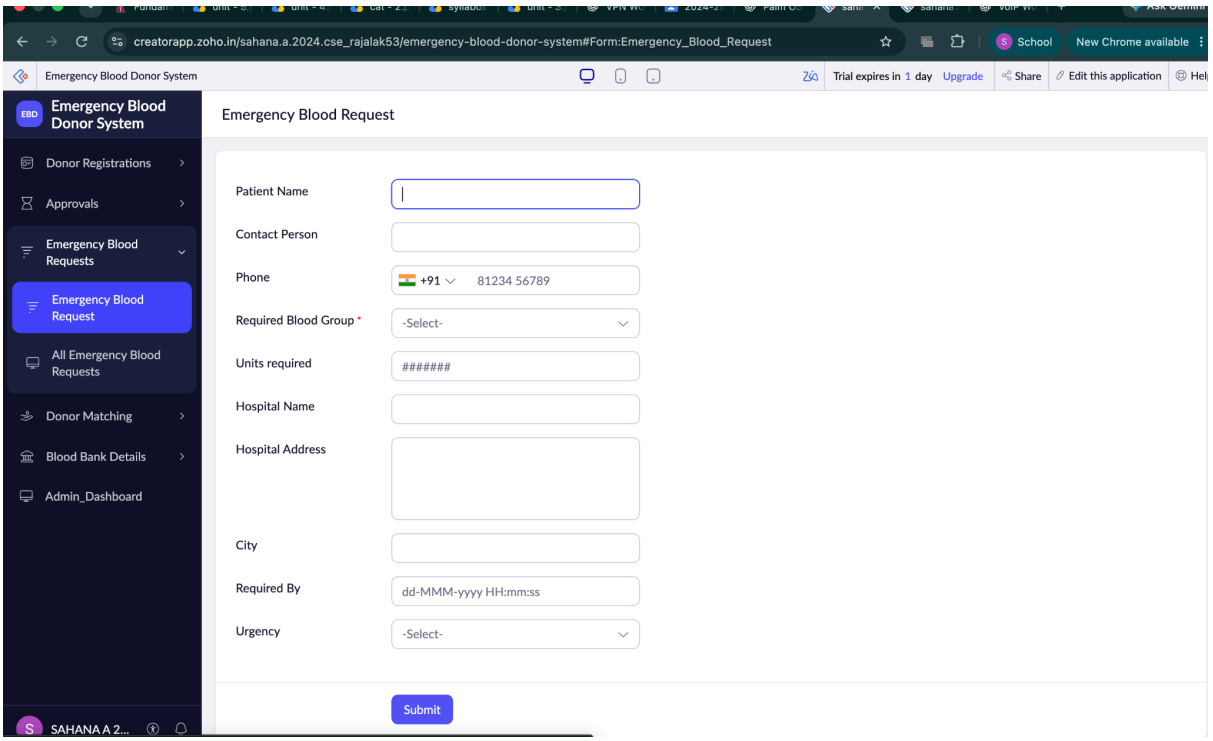


Figure 5.11: *Dashboard*

5.5 Deluge Scripting

Deluge is the scripting language of Zoho Creator [2]. In this project it is used to set default values, validate input, shortlist donors and send notifications. Table 5.4 summarises the scripts, and the sample code that follows is illustrative.

No.	Trigger / area	Logic summary
1	Blood_Request – on load	Set Request_Status to Pending.
2	Blood_Request – on validate	Reject a request whose Units_Required is not greater than zero.
3	Find Potential Donors (button or custom action)	Fetch available donors with the same blood group and city; create Donor_Match records without duplicates; set the request In Progress.
4	Donation_History – on success	Mark the related match as Donated and update the request status.
5	Notifications (optional)	Send an e-mail to the administrator or requester where configured.

Table 5.4: Deluge Logic Summary

Important: the scripts below are illustrative examples written for this report. The form and field link names are assumed and must be matched to the live application. No successful execution is claimed.

Sample 1 – Default request status (illustrative)

```
// Blood_Request > On Add > On Load
input.Request_Status = "Pending";
```

Sample 2 – Find Potential Donors (illustrative)

```
// Illustrative: reqID is the ID of the selected Blood_Request record
req = Blood_Request[ID == reqID];
donors = Donor_Registration[Blood_Group == req.Required_Blood_Group
```

```
    && City == req.City && Availability_Status == "Available"];
added = 0;
for each d in donors
{
    dup = Donor_Match[Request_ID == req.ID && Donor_ID == d.ID];
    if(dup.count() == 0)
    {
        insert into Donor_Match
        [
            Added_User = zoho.loginuser
            Request_ID = req.ID
            Donor_ID = d.ID
            Match_Status = "Potential"
        ];
        added = added + 1;
    }
}
if(added > 0)
{
    req.Request_Status = "In Progress";
}
```

Sample 3 – Validation and notification (illustrative)

```
// Blood_Request > On Add > On Validate
if(input.Units_Required <= 0)
{
    alert "Units required must be greater than zero.";
    cancel submit;
}

// Optional notification on success (sender and recipient to be configured)
sendmail
[
    from: zoho.adminuserid
    to: "[ADMIN EMAIL]"
    subject: "New blood request: " + input.Required_Blood_Group
    message: "A " + input.Request_Type + " request has been registered."
];
```

5.6 Workflows and Automation

Zoho Creator workflows run when a record is created, edited or submitted [1]. The proposed workflows are listed in Table 5.5. They have not been verified in this report; their configuration and execution history should be attached as evidence.

No.	Workflow	Form / event	Action
1	Donor registration	Donor_Registration – on add	Validate mandatory fields; default availability and date
2	Blood request creation	Blood_Request – on add	Validate input; record request
3	Default status	Blood_Request – on load	Set Request_Status = Pending
4	Find Potential Donors	Blood_Request – button	Match by blood group, city, availability
5	Status updates	Donor_Match / Blood_Request – on edit	Update match and request statuses
6	Donation history update	Donation_History – on success	Update match and request after a donation
7	Notifications	Blood_Request / Donor_Match	Send e-mail where configured

Table 5.5: Workflow Summary

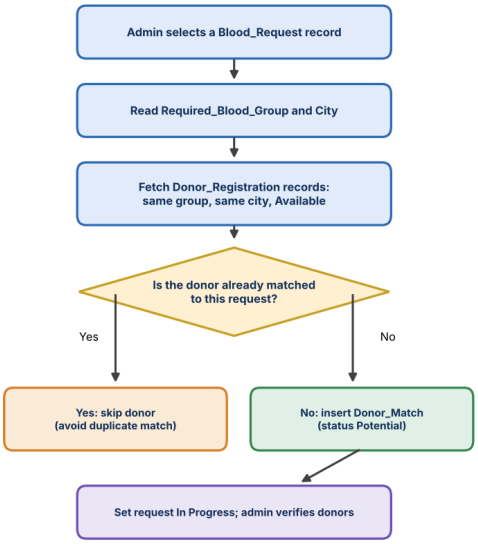


Figure 5.7: *Donor matching workflow*

The matching workflow reads the blood group and city of the selected request, fetches the available donors that match, skips donors who have already been matched to the same request and creates a Donor_Match record for each remaining donor. The request is then set to In Progress so that it is no longer listed as Pending.

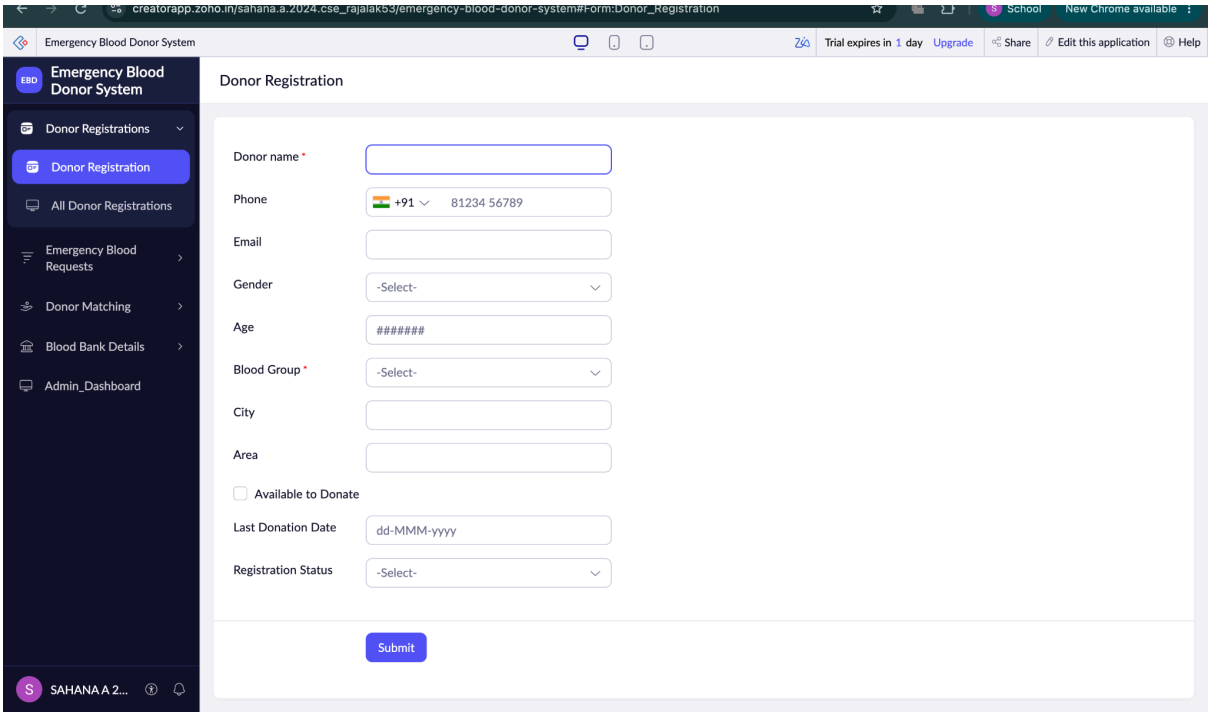


Figure 5.8: *Workflow configuration*

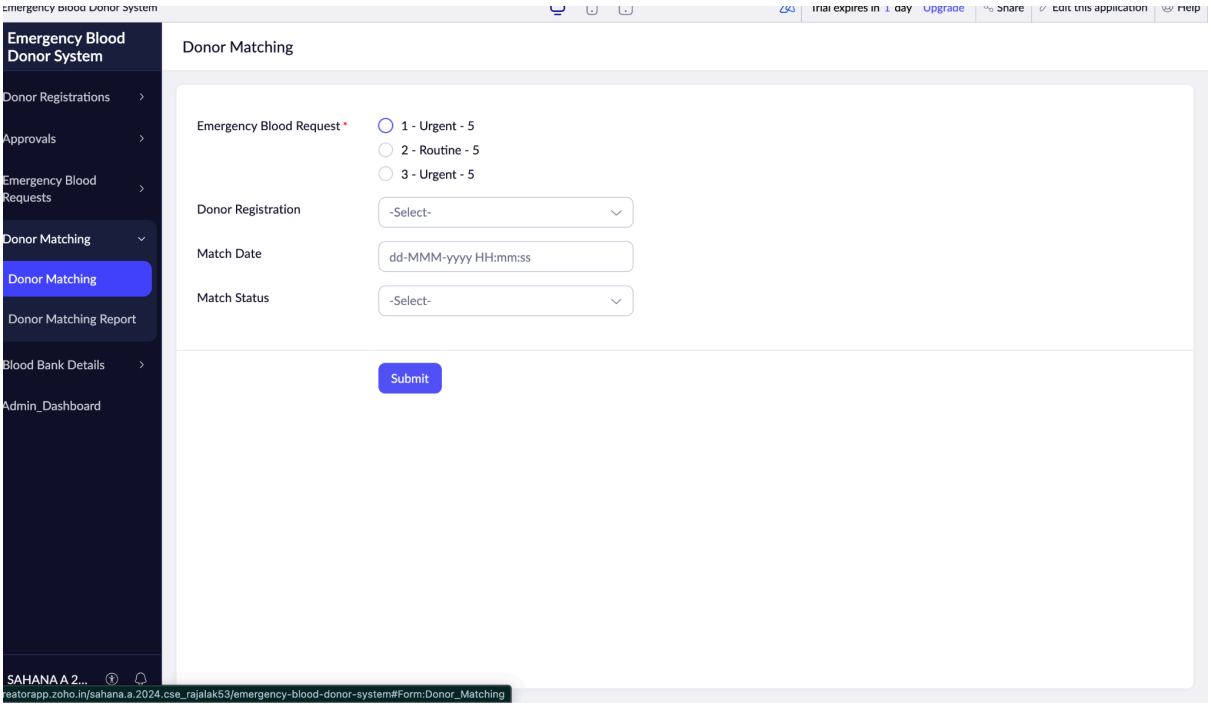


Figure 5.9: *Deluge script editor*

CHAPTER 6
TESTING AND VALIDATION

6.1 Testing Strategy

Testing should cover individual field rules, workflow triggers, linked-record consistency and end-to-end scenarios. Each test is performed with valid and invalid data and the result is recorded. The cases below are a test plan; **no actual results are reported**. The Actual Result column is to be completed after execution in the live application, and a case may be marked Pass only when supporting evidence exists.

6.2 Test Cases

Test ID	Scenario	Input	Expected result	Actual result	Status
TC-01	Donor registration	Valid donor details	Record saved with Donor ID and date	To be tested	Pending verification
TC-02	Required field validation	Submit donor form without blood group	Submission blocked with a message	To be tested	Pending verification
TC-03	Blood group selection	Open Blood Group drop-down	Eight blood groups are listed	To be tested	Pending verification
TC-04	Blood request submission	Valid request details	Request saved with status Pending	To be tested	Pending verification
TC-05	Invalid units	Units required = 0	Submission blocked	To be tested	Pending verification
TC-06	Emergency request	Request Type = Emergency	Request appears in Emergency Requests report	To be tested	Pending verification
TC-07	Hospital registration	Valid hospital details	Hospital appears in Hospital List	To be tested	Pending verification
TC-08	Donor matching	Run matching for an A+ request in a city with A+ donors	Matching donors listed in Donor_Match	To be tested	Pending verification
TC-09	Location filtering	Donor in a different city	Donor is not matched	To be tested	Pending verification
TC-10	Availability filtering	Donor marked Not Available	Donor is not matched	To be tested	Pending verification
TC-11	Duplicate match	Run matching twice for one request	No duplicate Donor_Match records	To be tested	Pending verification
TC-12	Request status update	Change status to In Progress, then Fulfilled	Status updates and reports reflect it	To be tested	Pending verification
TC-13	Donation history entry	Record a donation for a request	Record saved; linked request visible	To be tested	Pending verification
TC-14	Report generation	Open each report	Each report lists the expected records	To be tested	Pending verification
TC-15	Dashboard counters	Add records and refresh dashboard	Counters match the report counts	To be tested	Pending verification
TC-16	Access permissions	Donor tries to open admin report	Access is not available to the donor	To be tested	Pending verification

Table 6.1: Test Cases

6.3 Debugging Deluge Scripts

When a workflow does not behave as expected, the first step is to confirm the form event and when it runs. The developer then verifies the field link names, the lookup values and how empty values are handled. Temporary diagnostic messages may be used during development

and removed before use with real data, because donor details are personal data. The developer should check whether another workflow updates the same field and whether the same event can run more than once, so that duplicate matches or repeated e-mails are avoided. The execution history of each workflow shows the trigger, the record and the outcome of every run and is the first place to look.

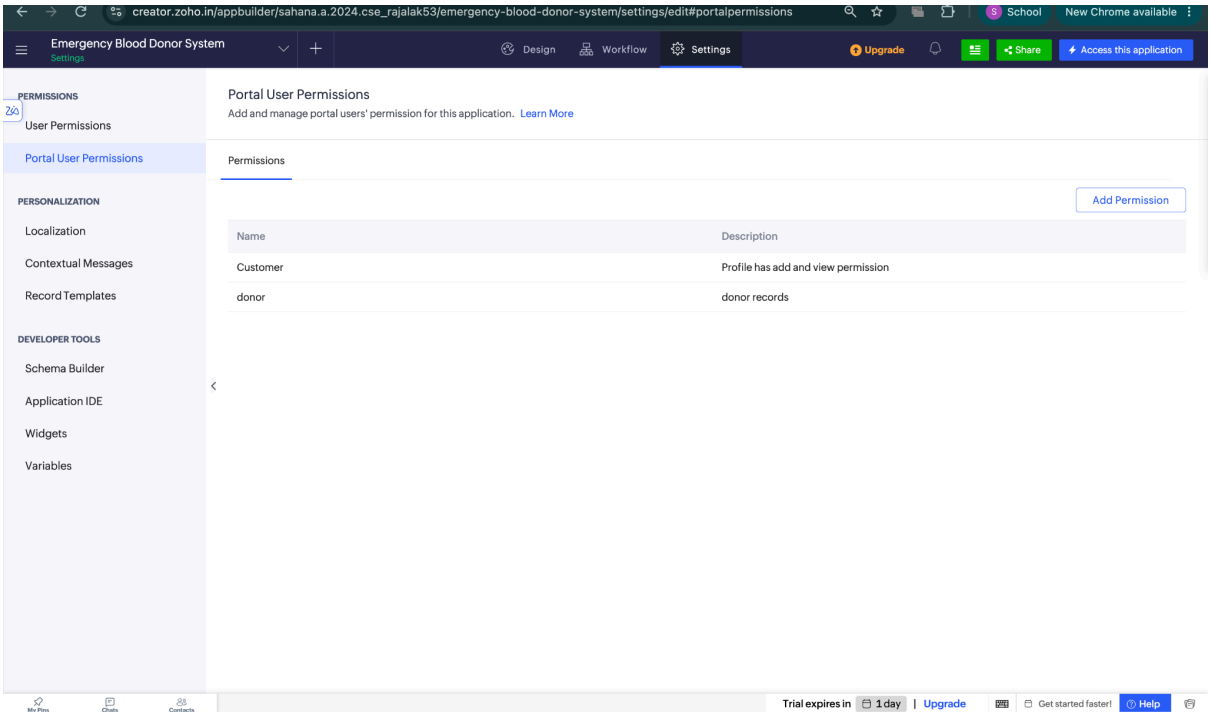


Figure 6.1: *Workflow execution history*

6.4 Validation and Risk Log

ID	Potential issue to monitor	Control	Status
R-01	Blood group or city typed differently in donor and request	Use drop-downs; standardise city names	Pending verification
R-02	Duplicate Donor_Match records	Check for existing match before inserting	Pending verification
R-03	Outdated availability status	Review and update availability regularly	Pending verification
R-04	Personal data exposed to unauthorised users	Restrict reports and fields by role	Pending verification
R-05	Matching mistaken for medical clearance	Display a verification notice; follow professional screening	Pending verification

Table 6.2: Validation and Risk Log

6.5 User Acceptance Testing

Acceptance testing should involve an administrator and a donor or requester following a complete scenario: register a donor, register a hospital, submit an emergency request, run donor matching, confirm that the expected donors are listed, update the statuses, record a donation and check the dashboard. Acceptance should be based on traceable records, correct counters, appropriate access and clear status changes. No signed acceptance sheet is available at the time of writing.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application should be developed and tested in a development environment, promoted to a stage environment and then published to production. A release checklist should confirm that forms, lookups, scripts, workflows, reports, the dashboard and permissions are configured in the target environment. This report does not include a published application URL or deployment record, so production deployment remains to be verified.

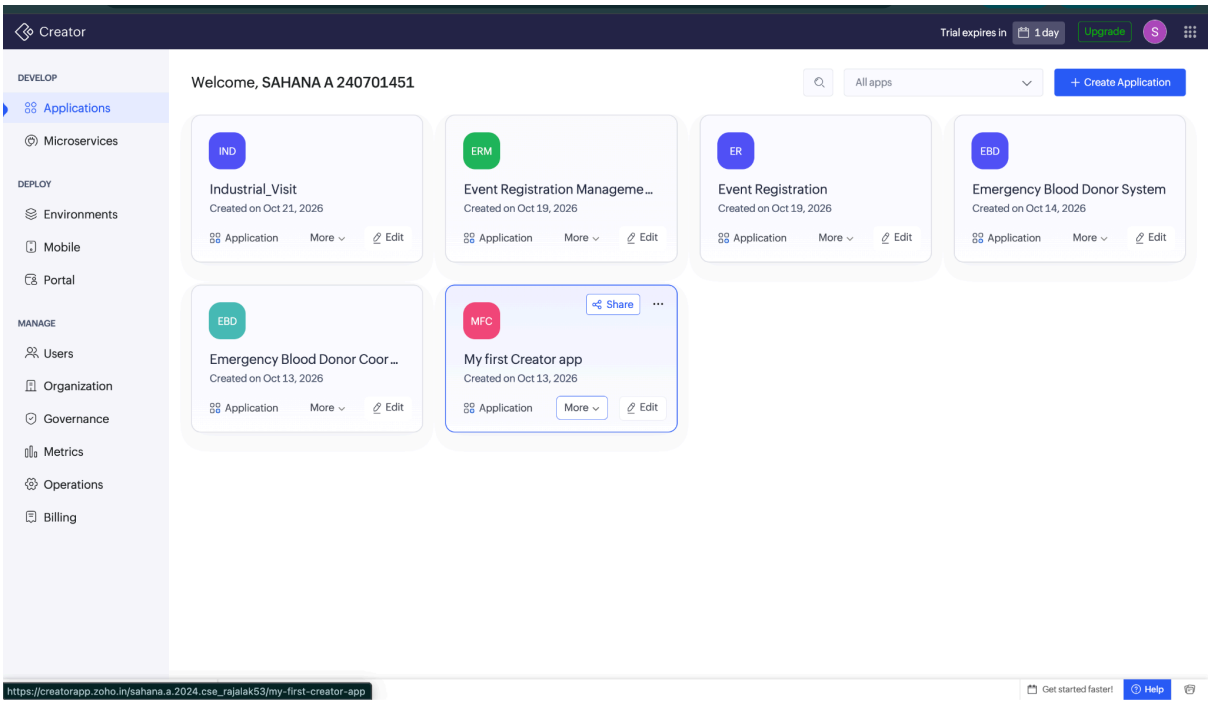


Figure 7.1: *Environments*

Deployment check	Verification
Data model	Confirm forms, field link names and lookups.
Master data	Enter hospitals and test donors with different blood groups and cities.
Automation	Activate and test workflows with sample records.
Notifications	Verify sender settings and recipient addresses.
Access	Test each role with least-privilege permissions.
Recovery	Define export, backup and change-control procedures.
Release	Record date, version, approver and known limitations.

Table 7.1: Deployment Checklist

7.2 Performance Optimisation

- Use report filters (for example Available Donors, Pending Requests) so that only relevant records are loaded.
- Fetch donors using criteria on blood group, city and availability instead of looping over all donors.

- Avoid repeated updates or repeated notifications in one workflow run.
- Keep field names and drop-down values consistent so that criteria are simple.

7.3 Security Features

Control	Application of the control (designed; to be verified)
Authentication	Users sign in to Zoho Creator or the portal before accessing the application.
Authorisation	Permissions limit which forms, reports and fields a role can use.
Privacy	Donor contact details are shown only to permitted users.
Validation	Mandatory fields, drop-down values and numeric checks.
Auditability	Requests, matches and donations are stored as separate records.
Error handling	Scripts should handle empty results and avoid exposing personal data in messages.

Table 7.2: Security Controls Designed for the Application

7.4 Governance and Administration

Governance covers who may change the configuration, how changes are tested and how the application is maintained. Because the portal stores personal information, the administrator should collect only the data that is needed, obtain the consent of donors to be contacted, correct or delete records on request and review access permissions periodically. Each release should be recorded with its date and version, and a short log of changes to workflows should be kept.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The application structure covers the blood request lifecycle from donor and hospital registration through request submission, donor matching, verification, donation recording and reporting. The expected scenario is described below; screenshots of the live application are to be inserted at the placeholders.

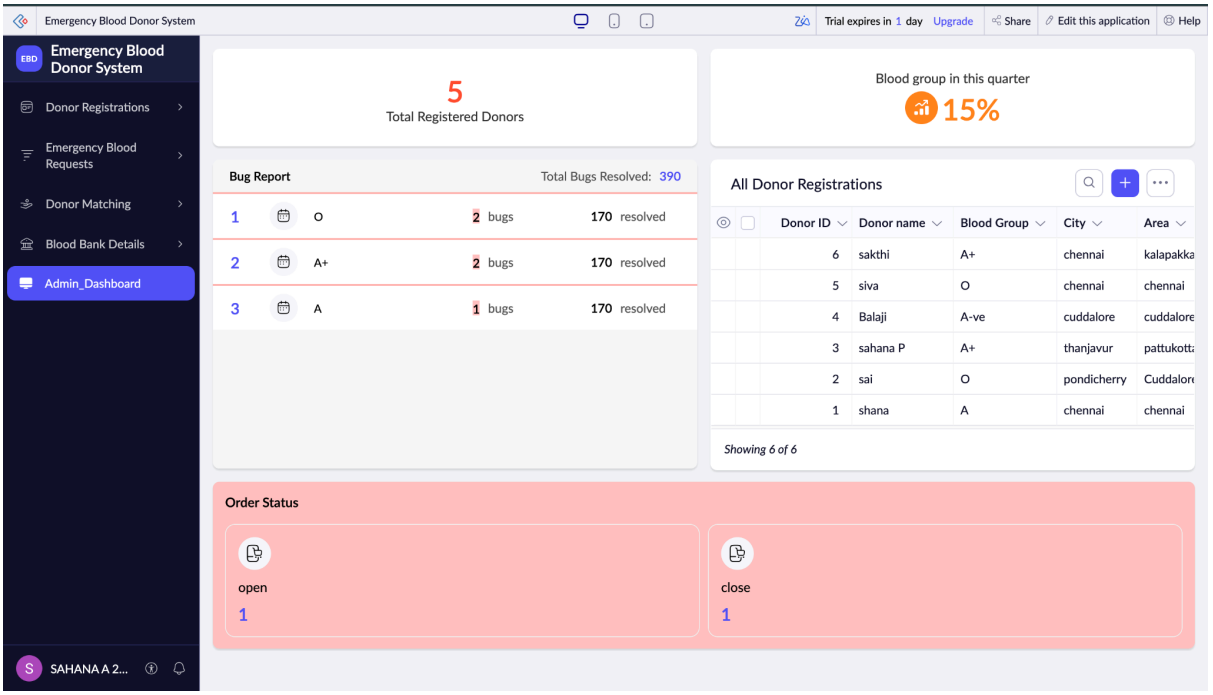


Figure 8.1: *Scenario step 1 – emergency blood request submitted*

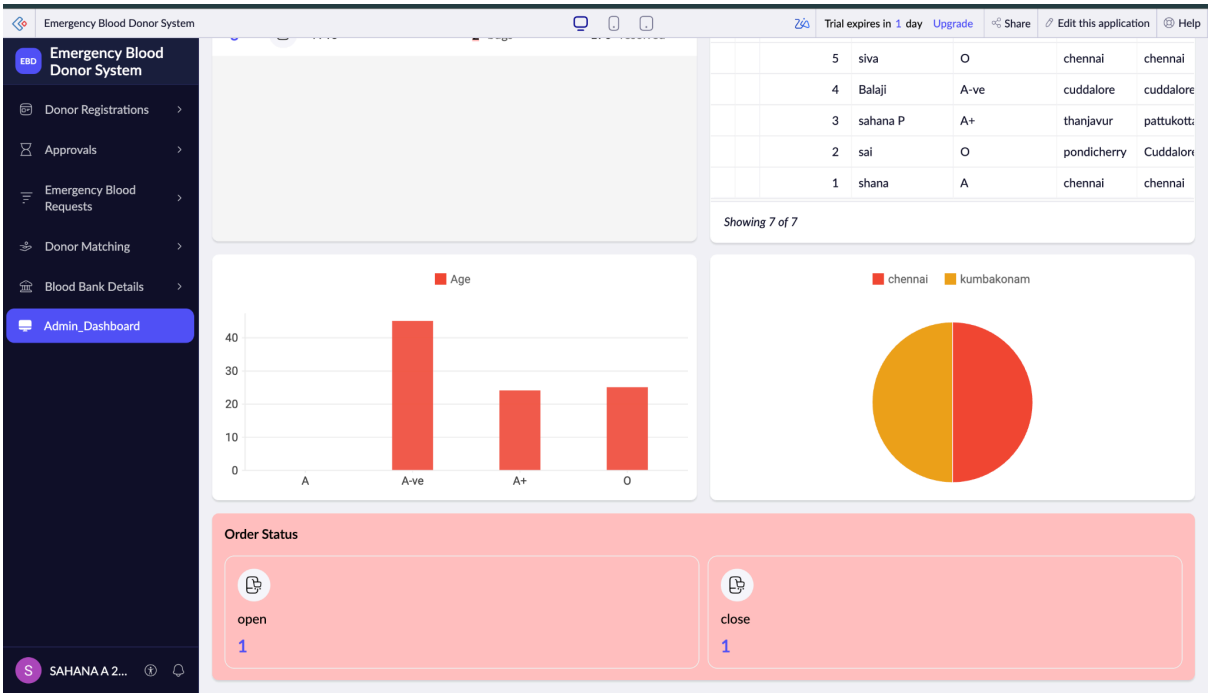


Figure 8.2: *Scenario step 2 – potential donors identified*

Feature group	Classification in this report	Evidence to attach
Five forms	Designed	Form screenshots with sample records
Default status Pending	Designed	Screenshot of a new request; test TC-04
Find Potential Donors	Designed (sample script)	Script and Donor_Match results; tests TC-08 to TC-11
Reports (8) and dashboard	Designed	Report and dashboard screenshots; tests TC-14, TC-15
Role-based access	Designed	Permission settings; test TC-16
SMS, mobile app, GPS search	Proposed for future	Not applicable

Table 8.1: Status Classification of Features

8.2 Achievement of Objectives

Objective	Design / deliverable	Assessment
Centralised donor database	Donor_Registration form and All Donors report	Covered in design; evidence required
Blood requests	Blood_Request form with type and status	Covered in design; evidence required
Hospital information	Hospital_Registration form and list	Covered in design; evidence required
Potential donor identification	Matching workflow and Donor_Match form	Requires execution evidence
Status tracking	Status fields and Pending Requests report	Requires test data
Donation history	Donation_History form and report	Covered in design; evidence required
Reduced manual coordination	Reports and dashboard	To be assessed by user trial

Table 8.2: Objective Achievement Mapping

8.3 Comparison with the Existing System and Advantages

Area	Manual process	Proposed portal
Donor data	Scattered in phones and groups	Centralised in Donor_Registration
Requests	Informal messages	Structured records with type and status
Finding donors	Phone calls to many people	Shortlist by blood group, city and availability
Tracking	No record of progress	Match and request statuses
History	Not maintained	Donation_History records
Visibility	Manual consolidation	Reports and dashboard

Table 8.3: Comparison with the Existing System

The advantages of the proposed portal are centralised donor information, easier management of blood requests, organised donor matching, improved accessibility of information, reduced manual coordination and better record maintenance.

8.4 Limitations

- The system depends on accurate donor information; wrong data gives wrong results.
- The system depends on internet connectivity and platform availability.
- Donor availability may change after the record was entered.
- Matching results are only a shortlist and must be confirmed by contacting the donor.
- Medical eligibility and blood compatibility must be verified by qualified professionals.
- Notification channels such as SMS need external services and configuration.
- Low-code does not remove the need for testing, access review and data-quality checks.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Visual form, report and dashboard builders reduce repetitive work, while Deluge gives control over matching and status rules. The project therefore demonstrates a low-code approach rather than a purely no-code solution. A pro-code service would be justified only for requirements that the platform cannot meet, such as advanced geospatial search.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Blood Donor System Portal presents a structured low-code solution for coordinating blood donors during emergencies. Its five forms connect donors, blood requests, hospitals, matches and donations. The workflow design addresses the main problems of the manual process by recording every request, shortlisting donors by blood group, location and availability, tracking statuses and keeping a donation history. The portal supports human coordinators; it does not replace medical screening.

The report follows the prescribed low-code project structure and documents the requirements, design, modules, testing approach and security considerations. Before the report is submitted as evidence of a working application, the test cases should be executed and the screenshots, workflow history and deployment details should be added.

9.2 Future Enhancements

- SMS and e-mail notifications to matched donors and requesters.
- A mobile application for donors and requesters.
- GPS-based donor search by distance from the hospital.
- Advanced analytics on request trends, response time and blood group demand.
- Integration with hospital and blood bank systems.
- Improved emergency request handling, such as automatic escalation of unanswered emergency requests.
- Reminders for donors based on their last donation date, as defined by medical guidelines.

These are proposed developments and are not part of the present work.

9.3 Learning Outcomes and Career Relevance

The project provides practical experience in requirement analysis, relational data modelling, form design, lookups, reports, dashboards, event-driven automation, Deluge scripting and test planning. These skills are relevant to roles in low-code development, business application development and process automation.

REFERENCES

[1] Zoho Corporation, “Zoho Creator Help Documentation,” Zoho. Available: https://www.zoho.com/creator/help/ [Accessed: date to be added].

[2] Zoho Corporation, “Deluge Help,” Zoho. Available: https://www.zoho.com/deluge/help/ [Accessed: date to be added].

[3] R. Elmasri and S. B. Navathe, Fundamentals of Database Systems. [Edition, publisher and year to be completed from the copy consulted.]

[4] I. Sommerville, Software Engineering. [Edition, publisher and year to be completed from the copy consulted.]

[5] R. S. Pressman, Software Engineering: A Practitioner's Approach. [Edition, publisher and year to be completed from the copy consulted.]

[6] MDN Web Docs, Mozilla. Available: https://developer.mozilla.org/ [Accessed: date to be added].

[7] OWASP Foundation, “Application Security Verification Standard (ASVS).” Available: https://owasp.org/www-project-application-security-verification-standard/ [Accessed: date to be added].

Note: only sources whose existence is certain are listed. Edition and year details for the textbooks, and access dates, must be completed by the student.

APPENDIX A
FEATURE STATUS CHECKLIST

Feature area	Status category	Evidence to attach / verify
Donor_Registration form	Designed	Form screenshot and sample records
Blood_Request form	Designed	Form screenshot; default status Pending
Hospital_Registration form	Designed	Form screenshot and sample hospitals
Donor_Match form	Designed	Form or report screenshot
Donation_History form	Designed	Form screenshot and sample donation
Matching workflow and script	Designed (illustrative script)	Script screenshot and execution history
Notifications	Designed / optional	Mail configuration and delivery check
Reports (8)	Designed	Report screenshots with sample data
Dashboard (7 counters)	Designed	Dashboard screenshot
Roles and permissions	Designed	Permission settings and test TC-16
Deployment	Not evidenced	Published URL, release date, checklist
Mobile app, SMS, GPS, analytics	Future	Not applicable

Table A.1: Feature Status Checklist

APPENDIX B
REPORTS AND PAGES CATALOGUE

B.1 Reports

- All Donors
- Available Donors
- All Blood Requests
- Pending Requests
- Emergency Requests
- Donor Matches
- Donation History
- Hospital List

B.2 Dashboard Indicators

- Total Donors, Available Donors
- Total Blood Requests, Pending Requests, Emergency Requests
- Successful Matches, Completed Donations

B.3 Routine Monitoring

Frequency	Monitoring activities
Daily	Review Emergency Requests and Pending Requests; contact potential donors.
Weekly	Review unmatched requests; update donor availability; review data quality.
Monthly	Review donation history, request fulfilment and permissions.

Table B.1: Routine Monitoring

APPENDIX C
DELUGE LOGIC SUMMARY

The following descriptions summarise the intended behaviour of the scripts. They are not deployable source code; the exact form and field link names must be taken from the live application.

C.1 Default request status

When the Blood_Request form loads, the Request_Status field is set to Pending.

C.2 Find Potential Donors

Read the blood group and city of the selected request. Fetch the Donor_Registration records with the same blood group and city whose Availability_Status is Available. For each donor, check whether a Donor_Match record already exists for this request; if not, insert one with the status Potential. If at least one record was created, set the request status to In Progress.

C.3 Donation recorded

After a Donation_History record is saved, set the related Donor_Match status to Donated and, when the units arranged are enough, set the Request_Status to Fulfilled.

C.4 Validation

On validate, check that mandatory fields are filled and that Units_Required is greater than zero; otherwise show an alert and cancel submission.

APPENDIX D
USER MANUAL

D.1 General Access

1. Open the authorised portal URL in a supported browser.
2. Sign in with the account provided.
3. Use the menu to open the form, report or dashboard for your role.
4. Fill every mandatory field, check the selections and submit.
5. Do not submit the same record again if a submission fails; check the message first.

D.2 Donor

1. Open Donor Registration and enter your details and blood group.
2. Set Availability Status to Available or Not Available.
3. Submit and update the status whenever your availability changes.

D.3 Requester

1. Open Blood Request and enter patient details, blood group, units and hospital.
2. Choose Normal or Emergency and submit.
3. The status is shown as Pending until the administrator acts.

D.4 Administrator

1. Register hospitals and review donor records.
2. Open the Pending Requests or Emergency Requests report and select a request.
3. Run Find Potential Donors, then contact and verify the donors.
4. Update match and request statuses.
5. Record each completed donation in Donation History.
6. Review the dashboard and reports regularly.

D.5 Troubleshooting

Issue	Recommended check
Record not submitted	Check mandatory fields and the units value.
No donors matched	Check blood group, city spelling and donor availability.
Duplicate matches	Check that the duplicate check in the script is active.
Counters look wrong	Check the criteria and the status values used.

Table D.1: Troubleshooting

APPENDIX E
EVALUATION RUBRIC

S.No.	Criterion	Indicators	Max. marks	Marks awarded
1	Problem definition and requirement analysis	Clarity of problem, objectives, FR/NFR, use cases	10	
2	System design	Architecture, ER model, form specifications, workflow design	10	
3	Forms, reports and pages	Field types, lookups, reports, dashboard	15	
4	Deluge scripting and workflows	Correctness and depth of scripts and workflows	15	
5	Integration, RBAC	Roles, permissions, integrations	10	
6	Testing, deployment, security	Test coverage, environments, security controls	10	
7	Live demonstration	End-to-end working across roles	15	
8	Project report	Template adherence, clarity, figures, references	5	
9	Viva-voce	Understanding of platform and design decisions	10	
	TOTAL		100	

Table E.1: Evaluation Rubric (to be completed by examiners)

Student: SAHANA A Register No.: 240701451 Date: _____

APPENDIX F
VIVA-VOCE RECORD SHEET

Q.No.	Question / topic asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

Table F.1: Viva-Voce Questions and Responses

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Examiners' remarks: _____

INTERNAL EXAMINER: _____ EXTERNAL EXAMINER: _____